Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 161 (2019) 851–858

Procedia Computer Science

www.elsevier.com/locate/procedia

The Fifth Information Systems International Conference 2019

The Influence of Discount Framing towards Brand Reputation and Brand Image on Purchase Intention and Actual Behaviour in e-commerce

Fanni Agmeka, Ruhmaya Nida Wathoni, Adhi Setyo Santoso*

President University, Cikarang 17550, Indonesia

Abstract

In e-commerce context, there are many implementations of discount framing that have effect toward brand reputation as well as brand image. The latter two constructs have been noted to affect consumer purchase intention and actual behaviour. However, there are limited studies that examine the relationship between discount framing consumer purchase intention and actual behaviour. Therefore, the objective of this study is to examine the impact of discount framing towards consumer's purchase intention and actual behaviour that may mediated by brand reputation as well as brand image. The researchers conducted a quantitative study with 307 valid sample data from the respondents in Greater Jakarta Area who have ever made purchases in Lazada, a leading e-commerce in Indonesia. The research results show that the path for discount framing in influencing purchase intention and actual behaviour depends on both brand reputation and brand image.

© 2019 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)
Peer-review under responsibility of the scientific committee of The Fifth Information Systems International Conference 2019.

Keywords: discount framing; brand reputation; brand image; purchase intention; actual behaviour

1. Introduction

The growth of e-commerce market in Indonesia had increased 17% in the past recent years that makes Indonesia become a potential market in digital business. This fact is important because it enables the seller to expand the trading

* Corresponding author. Tel.: +62-21-891-097-62.
E-mail address: adhi.setyo@president.ac.id

1877-0509 © 2019 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)
Peer-review under responsibility of the scientific committee of The Fifth Information Systems International Conference 2019.
10.1016/j.procs.2019.11.192

network sales [1]. Brand promotion through online advertising will also increase the brand awareness and brand image since the social media user can easily interact with other people. In addition, they can influence the others by providing positive testimonial [2].

One of e-commerce in Indonesia that maximizes online advertising is Lazada. In South East Asia, Lazada started their online marketplace by establishing their headquarters in Singapore and then expanded their business into other 5 Southeast Asia countries including Malaysia, Philippines, Thailand, Vietnam, and Indonesia. Lazada have been operated in Indonesia since 2012. In April-June 2017, Lazada is the top first performing online consumer goods retailer in Indonesia with 58.333.000 website visitor of Lazada in period April-June 2017 [3]. For supporting their growth, Lazada's seller often create discount for selected product promotion. Specifically, they make a discount framing by crossing out the original price and then make a new price above it with slightly lower amount. There are sellers who also increase the price of the certain product and then create a discount framing afterwards. Even, some of Lazada's sellers make un-logical price for the original one before they make discount framing. In this case, consumer will think in several times about e-commerce image and reputation first before making a decision to purchase the product [4]. Hence, it is important for the e-commerce firms to understand consumer purchase intention and actual behaviour determinants in order to keep the positive promotion effect.

In the digital marketing and consumer behaviour literature point of view, brand image and brand reputation were mentioned as the antecedents of purchase intention as well as actual behaviour [4]. However, the role of discount framing in this concept has not been examined yet [5]. In order to fill this research gap, the research objective of this study is to examine the effect of discount framing towards brand reputation and brand image that can influences on consumer purchase intention and actual behaviour afterwards. Thus, this study raises the following research questions to achieve the research objective: what is the role of discount framing in influencing consumer's purchase intention and actual behaviour?

2. Theoretical foundation

This section discusses the theoretical foundation of each variable used in this study. It includes discount framing, brand reputation, brand image, purchase intention, and actual behavior.

2.1. Discount framing

Discount framing is defined as a way to frame the price promotional in a certain offer [5]. Discount framing can also be defined as the manner by which the price of a particular product offers is framed in order to influences consumers' reaction [6]. Mostly, stores use discount framing to offer bundling products. When consumers see certain discount that show the crossed out nominal price and the new lower price as its comparison, the consumers will think about the opportunity for cost saving. Thus, creating a discount framing can increase consumer's purchase intention of any products. The previous study found that consumers do not really feel guilty when buy a product with a discount framing. Therefore, discount framing can increase the sales of the product [7].

2.2. Brand reputation

Brand reputation is the term to characterize the attitude of the consumer toward a good and reliable brand. It is defined as consumers' perception of a product quality that related to the brand name [8]. The consumers' perception itself also depends on consumers' social status when they perceive the products quality, whether they are reputable enough or not [9]. Therefore, brand reputation is important to strengthen consumer social status and their wealth. The consumers want to enhance their social self-actualization by having connection with a brand with have high reputation. When they choose highly reputable brands, other people will perceive their high life status. The researcher found that consumers who buy reputable brands are the consumers who want to show their social status. When consumers purchase any products, they will think about brand reputation. They perceive smaller risk toward brands with high reputation. Hence, it can stimulate buying decision [9].

Nowadays, the society materialism often motivates the consumption status to consumers who uses a reputable brand. Consumers think that consumption status is an important behaviour because other people perceive someone cultural belief when they see the brands that others choose to buy. Most of the people who choose to buy reputable brand want to improve their social status or prestige. Some of them have increase their income level so to enable them to use the luxury and reputable brand that can reflect their social position. Thus, the consumption status is related with the brand reputation that affected by the brand status and social class of good reputation and high quality [10].

2.3. Brand image

Brand image is defined as a consumer perception in their memory about brand that reflected as brand association [8]. Brand image is also considered as one of the most important intangible assets that have an impact on consumer perception in company [9]. There are two main components of brand image including functional and affective dimension. The functional dimension is a dimension that has measurable tangible characteristics. In contrast, the affective dimension is about psychological by feelings and attitudes toward the company [9]. The companies conduct both brand image development as well as brand portfolio expansion. By expanding their brand portfolio to a new product, they can get the advantages from the established brand image and awareness. Furthermore, it can also create more potential customers aware about their brand image [10].

2.4. Purchase intention

Purchase intention is an important variable that measure potential action that consumer choose to be done [11]. By understanding consumers' purchase intention, it can help the companies to understand the market and adjust the offered product or services that enable them get more sales and earn profit [11]. Moreover, understanding consumers' purchase intention can predict the customer's retention of a particular brand [12].

2.5. Actual behaviour

Actual behaviour defines as consumer behaviour affected by their intention to act and influenced by their attitude [13]. Furthermore, actual behaviour refers to the way the people make a decision to spend their time, money and effort on consuming products or services [13]. Actual behaviour is related with Theory of Reasoned Action (TRA) that explains about people perception regarding others opinion that is affected people intention and attitude. TRA is also useful in predicting the behaviour. For example, this theory was applied in various fields such as public behavioural intention in Uganda, the intention that architect take in a design team, and also the intention to choose halal product [14]. The main aim of TRA is to determine consumer behaviour to engage or not engage their intention [14]. Human behaviour is determined by behavioural and antecedent intentions such as attitudes toward subjective behaviour and norms. Because of these three determinants, this study uses this theory [14].

2.6. Hypothesis development

This sub-section presents hypothesis development of this study based on the latent variables' theoretical foundation discussed previously as well as the findings in the previous study. The developed hypothesis is constructed in certain manner to be tested with quantitative analysis. Then, the results of quantitative analysis answer the research question raised in the introduction section.

In the previous literature, there is a different discount framing effectiveness related with the product original price. For the expensive product, discount framing is more effective when consumers see the crossed out original price and discounted price as the comparison. In contrasts, for the cheaper product, showing the percentage of price reduction is more effective [15]. The findings in this study show that when consumer see a discount framing with both amount of the price and the percentage of price reduction, they will see the brand name first and then think about the store brand reputation and image. Hence, this study suggests that discount framing influence brand reputation and brand image and formalize them into Hypothesis 1 and Hypothesis 2 as follow:

- *Hypothesis 1: Discount framing has positive influence on brand reputation*
- *Hypothesis 2: Discount framing has positive influence on brand image*

Brand reputation can influence the brand image. For example, the product with good brand image can turn into bad image product from the consumer perception when there are some moments that give bad reputation, and vice versa. Brand reputation also has role to help consumers to make choices and also to express the value of a particular brand [16]. The previous study mention that Ferrari is the most reputable car brand in the world and have positive brand image [16]. Thus, the researcher suggests that brand reputation influence brand image as formalized into Hypothesis 3 as follow:

- *Hypothesis 3: Brand reputation has a positive influence on brand image*

The highly competitive industry such as restaurant should have high brand reputation to create consumer trust as well as to reduce the uncertainty [17]. Consumer trust is a factor of buying behaviour that can reduce the consumers perceive uncertainty and risk. Furthermore, consumers believe that the food quality is good when they trust the brand. It makes the reputable brand influence their buying behaviour. In other words, when the brand reputation of the store is high, the possibility for them to purchase the product is higher, and vice versa. Consumer's pride also can increase when they use reputed brand. The previous research found that more people prefer to buy reputable brand to increase their pride [17]. This study context with e-commerce industry is also highly competitive industry. Thus, this study suggests that brand reputation influence purchase intention as formalized into the following Hypothesis 4:

- *Hypothesis 4: Brand reputation has a positive influence on purchase intention*

In restaurant industry the competition is very tight. It makes the restaurant should pay attention to their branding. Since the consumer lifestyle is constantly changing, consumers prefer to buy food from the branded restaurants that have strong personalities and identities. Moreover, they even are willing to pay more to be more considered by other people when they make a purchase in branded restaurant [17]. The study in Europe and USA found that a store that has high brand image tend to have greater margin because brand image can make consumer perception to purchase or not purchase the goods [10]. In addition, brand image is considered as an important aspect when the consumers want to make a purchasing decision for the goods. A store that has high brand image can increase consumer's intention to purchase the goods [9]. Hence, this study suggests that brand image will influence consumer's intention to purchase the goods as formalized into the following Hypothesis 5

- *Hypothesis 5: Brand image has positive influence on purchase intention*

There are various impacts of purchase intention toward consumer actual behaviour related with the consumers' background. For example, the purchase intention from consumer with 'green' attitude impacts his or her action in purchasing green product [18, 19]. For social media user, the consumer the impact of attitudes toward sponsored recommendation posts on actual behaviour in purchasing the recommended product is significantly positive [20].

In Theory of Reasoned Action (TRA) perspective, the potential risk that consumer perceives has direct negative effect to the actual behaviour in shopping online [14]. Thus, this research uses this theory to predict the consimer actual behaviour when consumer sees discount framing offered by the e-commerce [20]. Based on this conceptualization, the Hypothesis 6 is formalized as follow:

- *Hypothesis 6: Purchase intention has positive influence on actual behaviour*

3. Methodology

3.1. Context

The context of this research focuses on recent discount framing practices in an emerging market, especially in Indonesia. Price and promotion are the important things when consumer wants to buy a product because it can affect their purchasing decision. Sometimes, consumer purchases the product because they are attracted by the promotion that the e-commerce offers [21]. Since many e-commerce nowadays attract consumer by provide vouchers, create discount framing, and also provide free delivery fee, the competition among the e-commerce becomes very tight [3]. In addition, discount framing created by the e-commerce enable the consumers have ambiguous perception [22].

3.2. Data collection and sample

The sample data was collected from Lazada customers in Greater Jakarta Area since this segment provides the highest contribution of e-commerce sales. A purposive sampling was used from a population with specific criteria; the respondents must have made any purchase in Lazada at least three times. This study measured a non-probability sampling technique, which was purposive in order to collect the data of respondents, which separated by using online platform called Google Docs. The data collection was conducted within 3 months from September 2017 until the end of November 2017. As a result, there are 350 online responses from the respondents. After we check the response data, there are 307 valid responses that can be used for research analysis.

3.3. Measurement

The measurement indicators of each variable was adapted and assembled from previous related research. In the face validity process, the researcher ask five people who are familiar with e-commerce purchase transaction to explain their perception about the questioner. This questionnaire use two different languages, Bahasa and English. The questionnaire was measured by using Seven-Point Likert scale to be a decent balance between positive and negative response. The scales are ranging from strongly disagree, disagree, somewhat disagree, neither agree or disagree, somewhat agree, agree, and to strongly agree. Each of latent variable in this study use the questionnaire as measurement indicators adapted from previous study presented in Table 1.

To test the validity of the data this research used factor analysis to measure the variable of the theoretical framework and identify which data were valid or invalid. The researcher evaluates the Kaiser-Meyer-Olkin (KMO) with value criteria greater than 0.5 and followed by Bartlett's test with value criteria lower than 0.05 as to determine the significance of the validity. The value criteria of Anti-Images Matrices should be greater than 0.5 as well for the communalities of the data, the percentage of Total Variance Explained should greater or equal than 60%, and the value criteria of Component Matrix should be greater than 0.5. After the validity test was finished, the reliability test was conducted by using all valid. In order to determine the reliability, the data should be fulfilling the value criteria of Cronbach's alpha (α) that should be greater than 0.6.

Table 1. Measurement Indicators.

Construct	Measurement Indicators	Adapted Source
Discount Framing	- I shop from e-commerce that give discount to the product that they sell.	[15]
	- I shop from e-commerce that not give discount to the product that they sell) I shop from e-commerce that not give discount to the product that they sell.	
	- I shop from e-commerce because the store gives cheaper price rather than the physical store's price.	
	- I shop from e-commerce that offers affordable prices.	
Brand Reputation	- Lazada is trustworthy.	[23]
	- Lazada is reputable.	

Construct	Measurement Indicators	Adapted Source
	- Lazada makes honest claim.	
	- Lazada is reliable.	
	- I know how Lazada looks like.	
Brand Image	- Lazada has differentiated image in comparison with the other e-commerce.	[24]
	- Being able to order by phone makes me interested to use P2P online transportation, since I am accustomed to making phone calls.	
	- Lazada has a clean image.	
	- Lazada is a well-established e-commerce.	
	- Lazada arouses my sympathy to purchase the product.	
Purchase Intention	- I have intention to buy product from Lazada.	[20]; [21]; [15]
	- I would recommend Lazada if a friend calls me to get my advice in his/her search for online shopping.	
	- It is likely that I will visit Lazada when doing online shopping.	
	- I have found online purchasing products from Lazada to be worthwhile.	
	- I intend to keep purchasing product by online through Lazada.	
Actual Behaviour	- I often buy product from Lazada.	[17]
	- I often buy products from Lazada because it is very easy to do.	
	- I often buy product from Lazada because they are more convenient.	
	- I often buy product from Lazada for my need.	
	- I often buy products from Lazada because it is safe to use.	

This study used Structural Equation Modelling (SEM) with SPSS Amos 24th version to do the inferential analysis and test the hypothesis. In order to test the hypothesis, this study use p-value criteria with 95% confidence level that should be less or equal than 0.05 for the significance result. There are some value criteria to indicates the Model Fit of data in SEM which consists of Goodness of Fit Indices, Adjusted Goodness of Fit Index, Comparative Fit Index, Incremental Fit Index, and Tucker Lewis Index (GFI, AGFI, CFI, IFI, and TLI > 0.90), (CMIN value < 5), Root Mean Square Error of Approximation (RMSEA) ($0.05 \leq RMSEA \leq 0.08$) [25]. The model fit represents about the fitness of the data collected from the survey with the theoretical model. This research also used Squared Multiple Correlation method with R-square test to measure and explain at what extent the exogenous variables can explain as the determinant of endogenous variables.

4. Research findings and discussion

This study used 37 respondent data for conducting the data analysis. The results of data analysis shows that the value of KMO-MSA Barlett's Test of all variables passes the criteria of validity test (≥ 0.5), the value of Cronbach Alpha passes the criteria of reliability test (>0.6), and the value of Goodness of Fit is fulfil the criteria mentioned in the previous section. These results show that the measurement indicators for all of the latent variables are valid and reliable to be used for inferential analysis. Furthermore, the Goodness of Fit result show that the collected data is fit with the theoretical model. Fig. 1 presents the research framework with hypothesis testing result.

The result for the first hypothesis that states that discount framing influences brand reputation is accepted ($p<0.000$). The previous research also supports this statement and mention that a marketer can build decent brand reputation by using discount framing. Discount framing made by the marketer is intended to attract the consumer to buy a particular brand [25]. The second hypothesis suggests that discount framing influences brand image. This hypothesis is rejected ($p=0.847$). The finding is also supported with the previous study. Sometimes people think about negative effect of a promotion made by the brand [26]. The third hypothesis that suggests brand reputation influences

brand image is accepted (p<0.000). It is also supported by previous study that mention about the damage of brand reputation can directly influences the brand image [27].

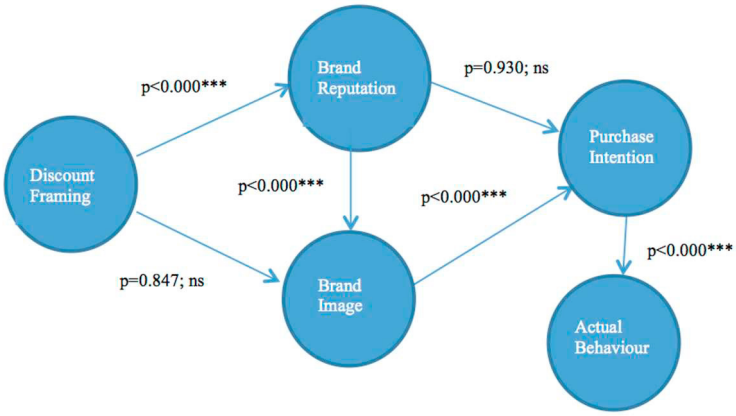

Fig. 1. Research Framework (with p-value for Hypothesis testing).

The fourth hypothesis suggests that brand reputation influences purchase intention is rejected (p=0.930). There was no significant influence of negative reputation towards purchase intention [23]. Next, the fifth hypothesis suggests that brand image influences purchase intention. The result is this accepted (p<0.000). It represents that brand image significantly influences purchase intention. The previous study mentions that decent brand image make the consumers have more intention to purchase the product [21, 28]. Finally, the last hypothesis is also accepted (p<0.000). It suggests that purchase intention influences actual behaviour. It also supported by the previous study that states about the intention to purchase a certain product has positive and significant effect toward the actual buying behaviour [17]. It means the consumers who have intention to buy the products can exhibit actual buying behaviour of buying the products.

5. Conclusion and recommendations

The results of the quantitative study approach among 307 respondents conclude that there is significant influence between the stated hypotheses above. Since some of the e-commerce companies make discount framing to attract the consumer to buy a product, this study examines the effect of discount framing towards consumers' purchase intention and actual behaviour. This study demonstrates the main issues of the willingness to purchase after seeing discount framing that the e-commerce offer. E-commerce can gain more advantages from this study by understanding the influential factors, brand reputation and brand image as the consequences path of discount framing, that lead the consumer to purchase the product. Since this study investigated the influential factors of consumer purchase intention of discount framing in e-commerce industry in Indonesia, future research can replicate the study in more than one service industry and set further analysis to generalize the model.

References

[1] Haidar, A. (2016) *Strategic Role of E-Commerce and MSMEs in MEA.* Available from: http://www.kompasiana.com/alvin123/peran-strategis-e-commerce-dan-umkm-dalam-mea_5861b5d961afbdea0ef3ecc8. [Accessed 27th December 2017].

[2] Tsimonis, Georgios, and Sergios Dimitriadis. (2014) "Brand Strategies in Social Media." *Marketing Intelligence & Planning* **(32) 3**: 328-344.

[3] Anggraeni, L. (2017) *E-Commerce Market Competition in Indonesia.* Available from: https://www.medcom.id/teknologi/news-teknologi/nN9VmM8b-persaingan-e-commerce-di-indonesia-kian-memanas. [Accessed 15th Decemer 2017].

[4] Jalilvand, Reza, Mohammad, and Neda Samiei. (2012) "The Effect of Electronic Word of Mouth on Brand Image and Purchase Intention: An Empirical Study in The Automobile Industry in Iran." *Marketing Intelligence & Planning* **30(4)**: 460-476.

[5] Janiszewski, C., and M. Cunha Jr. (2004). "The Influence of Price Discount Framing on The Evaluation of A Product Bundle." *Journal of Consumer Research* **30 (4)**: 534-546.
[6] Gendall, Philip, Janet Hoek, and Tracy Pope. (2006) "Message Framing Effects on Price Discounting." *Journal of Product & Brand Management* **15 (7)**: 458-465.
[7] Khan, Uzma, and Ravi Dhar. (2010) "Price-framing Effects on The Purchase of Hedonic and Utilitarian Bundles." *Journal of Marketing Research* **47 (6)**: 1090-1099.
[8] Rindell, Anne, and Oriol Iglesias. (2014) "Context and Time in Brand Image Constructions." *Journal of Organizational Change Management* **27 (5)**: 756-768.
[9] Martínez, Patricia, Andrea Pérez, and Ignacio Rodríguez Del Bosque. (2014) "CSR Influence on Hotel Brand Image and Loyalty." *Academia Revista Latinoamericana de Administración* **27(2)**: 267-283.
[10] Kremer, Florence, and Catherine Viot. (2012) "How Store Brands Build Retailer Brand Image." *International Journal of Retail & Distribution Management* **40 (7)**: 528-543.
[11] Chen, Mei-Fang, and Chia-Lin Lee. (2015) "The Impacts of Green Claims on Coffee Consumers' Purchase Intention." *British Food Journal,* **117 (1)**: 195-209.
[12] Kudeshia, Chetna, and Amresh Kumar. (2017) "Social eWOM: Does it Affect The Brand Attitude and Purchase Intention of Brands?." *Management Research Review* **40 (3)**: 310-330.
[13] Liu, Yu, Sigal Segev, and Maria Elena Villar. (2017) "Comparing Two Mechanisms for Green Consumption: Cognitive-Affect Behavior vs Theory of Reasoned Action." *Journal of Consumer Marketing* **34 (5)**: 442-454.
[14] Lujja, S., M. Omar Mohammad, and R. Hassan. (2016) "Modelling Public Behavioral Intention to Adopt Islamic Banking in Uganda: The Theory of Reasoned Action." *International Journal of Islamic and Middle Eastern Finance and Management* **9 (4)**: 583-600.
[15] McKechnie, Sally, James Devlin, Christine Ennew, and Andrew Smith. (2012) "Effects of Discount Framing in Comparative Price Advertising." *European Journal of Marketing* **46 (11/12)**: 1501-1522.
[16] Lowenstein, Michael W. (2011) "The Customer Advocate and The Customer Saboteur: Linking Social Word-of-Mouth, Brand Impression, and Stakeholder Behavior." *Wisconsin, ASQ Quality Press*.
[17] Wee, Chiew Shi, Mohd Shoki Bin Md. Ariff, Norhayati Zakuan, Muhammad Naquib Mohd Tajudin, Khalid Ismail, and Nawawi Ishak. (2014) "Consumers Perception, Purchase Intention and Actual Purchase Behavior of Organic Food Products." *Review of Integrative Business and Economics Research,* **3 (2)**: 378-397.
[18] Doszhanov, Aibek, and Zainal Ariffin Ahmad. (2015) "Customers' Intention to Use Green Products: The Impact of Green Brand Dimensions and Green Perceived Value." in *SHS Web of Conferences*, *EDP Sciences* **18**:1-16.
[19] Rizwan, Muhammad, Usman Mahmood, Hammad Siddiqui, and Arham Tahir. (2014) "An Empirical Study About Green Purchase Intentions." *Journal of Sociological Research* **5 (11)**: 948-5468.
[20] Lu, Long-Chuan, Wen-Pin Chang, and Hsiu-Hua Chang. (2014) "Consumer Attitudes Toward Blogger's Sponsored Recommendations and Purchase Intention: The Effect of Sponsorship Type, Product Type, and Brand Awareness." *Computers in Human Behavior* **34**: 258-266.
[21] Yee, Cheng Pau, Kenny Wee Khai Mun, Leow May Yee, and Yeo Ling. (2014) "Factors Influencing Purchase Intention of Starbucks." *Dissertation. UTAR.*
[22] Choi, Sungchul, Xin Ge, and Paul R. Messinger. (2010) "Consumer Perceptions of Ambiguous Price Promotions: Scratch and Save Promotions Versus Tensile Price Claims." *Journal of Product & Brand Management* **19 (7)**: 477-486.
[23] Waddington, Andrew John. (2011) "Investigating The Impact of Brand Reputation on Brand Architecture Strategies: A Study on A South African Automotive Company." *Doctoral Dissertation. Rhodes University.*
[24] Jalilvand, Reza, Mohammad, and Neda Samiei. (2012) "The Effect of Electronic Word of Mouth on Brand Image and Purchase Intention: An Empirical Study in The Automobile Industry in Iran." *Marketing Intelligence & Planning* **(30) 4**: 460-476.
[25] Yen, Po-Yin, Karen H. Sousa, and Suzanne Bakken. (2014) "Examining Construct and Predictive Validity of The Health-IT Usability Evaluation Scale: Confirmatory Factor Analysis and Structural Equation Modeling Results." *Journal of the American Medical Informatics Association* **21 (e2)**: e241-e248.
[26] Chowdhury, Sultana R. (2017) "Building Brand Reputation through Marketing Mix-A Case Study on Philips and Toshiba." *European Journal of Business and Management* **9 (8)**: 11-16.
[27] Waanders, Kooen. (2013) "The Effectiveness and Consequences of Price Promotions." [master thesis], *University of Twente.*
[28] Jung, Na Young, and Yoo-Kyoung Seock. (2016) "The Impact of Corporate Reputation on Brand Attitude and Purchase Intention." *Fashion and Textiles* **(3) 1**: 20-34.